Why spaced retrieval works

A four-page summary that comes with Etude's Notes tutorial. Open the ≡ panel to jump between its sections.

1. The forgetting curve

Right after you learn something you can recall it easily. Then the chance that you still can falls, fast at first and slower later. Ebbinghaus measured this on himself in the 1880s, and it has been reproduced many times since.

Schedulers like FSRS model that chance as a curve that depends on how long it has been since your last review and on how *stable* the memory is. In FSRS-6 the curve is

$$R(t) = (1 + 19/81 \cdot t / S)^{-1/2}$$

where `t` is the number of days since the review and `S`, the stability, is the number of days after which the chance drops to 90%. Every successful review makes `S` larger, so the curve flattens and the next review can wait longer.

2. Retrieval practice

Reading your notes again feels productive, but it mostly makes them familiar. Pulling an answer out of memory, without looking, changes the memory itself: it gets easier to find the next time. This is called the *testing effect*, and it holds even when nobody grades you.

> Write the note, then make a card that asks for what the note says. Answer the card instead of rereading the note.

Retrieval also tells you what you don't know. A card you get wrong points straight back at the page you need to read again, which is why Etude keeps your cards and your notes side by side.

3. Spacing

The same amount of practice works better spread out than crammed. Ten minutes a day for a week beats seventy minutes the night before, especially for what you still want to know a month later.

The best moment to review is just before you would have forgotten: early enough that you can still recall it, late enough that the recall takes some effort. A scheduler picks that moment for each card from its history, so you don't have to.

- Short gaps while a card is new.
- Longer and longer gaps once you know it.
- A short gap again after you miss it.

4. Mixing and effort

Practising one kind of problem at a time feels smooth, but mixing kinds (*interleaving*) teaches you to tell them apart, which is most of what an exam asks. Difficulties like this, which slow you down now but help later, are sometimes called *desirable difficulties*.

Recalling some things can even make related things harder to recall for a while (*retrieval-induced forgetting*). Reviewing a topic as a whole, not just its favourite parts, keeps the neighbours alive too.

Further reading

H. Ebbinghaus (1885). *Über das Gedächtnis.*

H. L. Roediger & J. D. Karpicke (2006). Test-enhanced learning. *Psychological Science* 17(3).

N. J. Cepeda et al. (2006). Distributed practice in verbal recall tasks. *Psychological Bulletin* 132(3).

M. C. Anderson, R. A. Bjork & E. L. Bjork (1994). Remembering can cause forgetting. *JEP: Learning, Memory, and Cognition* 20(5).

J. Ye. The FSRS algorithm, github.com/open-spaced-repetition.